

YULIN LUO

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👤 RESEARCH PROFILE

I am a Ph.D. candidate at Peking University, advised by Prof. Shanghang Zhang. My research focuses on open-world generalization for embodied intelligence, with emphasis on generalizable embodied foundation models, robot-scenario data and benchmark construction, and model-data co-evolution.

🎓 EDUCATION

Peking University, Ph.D. Candidate Sep. 2023 – Present

- Advisor: Prof. Shanghang Zhang.

Shanghai Jiao Tong University, B.Eng. in Automation Sep. 2019 – Jun. 2023

- GPA: 3.87/4.30; Major rank: 2/89.

🔬 RESEARCH INTERESTS

- **Model:** building generalizable embodied foundation models for open-world robot intelligence.
- **Data / Benchmark:** constructing embodied-agent benchmarks, multimodal evaluations, and robot-scenario data.
- **Model-Data Co-evolution:** scaling embodied intelligence through data-driven model improvement and model-guided data construction.

📄 SELECTED PUBLICATIONS

Timeline: 2022 MICCAI · 2024 ECCV/AAAI · 2025 RSS · 2026 ACM MM/ECCV/AAAI Oral

1. RoboBench: A Comprehensive Evaluation Benchmark for MLLMs as Embodied Brain
Yulin Luo^{*}, Chun-Kai Fan^{*}, Menghang Dong^{*}, Jiayu Shi^{*}, Xiangju Mi^{*}, et al. *ECCV'26 (CCF-B)*
2. LLM as Dataset Analyst: Subpopulation Structure Discovery with Large Language Model
Yulin Luo^{*}, Ruichuan An^{*}, Bocheng Zou, Yiming Tang, Jiaming Liu, and Shanghang Zhang. *ECCV'24 (CCF-B)*
3. Look Before Acting: Enhancing Vision Foundation Representations for Vision-Language-Action Models
Yulin Luo^{*}, Hao Chen^{*}, Zhuangzhe Wu^{*}, Bowen Sui^{*}, Jiaming Liu^{*}, et al. *ACM MM'26 (CCF-A)*
4. Decomposing the Neurons: Activation Sparsity via Mixture of Experts for Continual Test-Time Adaptation
Rongyu Zhang^{*}, Aosong Cheng^{*}, **Yulin Luo**^{*}, Gaole Dai, Huanrui Yang, et al. *AAAI'26 Oral (CCF-A)*
5. RoboMIND: Benchmark on Multi-Embodiment Intelligence Normative Data for Robot Manipulation
Kun Wu, Chengkai Hou, Jiaming Liu, et al., **Yulin Luo**, et al. *RSS'25 (CCF-B)*
6. RandStainNA: Learning Stain-Agnostic Features from Histology Slides by Bridging Stain Augmentation and Normalization
Yiqing Shen^{*}, **Yulin Luo**^{*}, Dinggang Shen, and Jing Ke. *MICCAI'22 (CCF-B)*
7. Efficient Deweather Mixture-of-Experts with Uncertainty-Aware Feature-Wise Linear Modulation
Rongyu Zhang, **Yulin Luo**, Jiaming Liu, Huanrui Yang, et al. *AAAI'24 (CCF-A)*

^{*} Equal contribution.

📁 RESEARCH INTERNSHIPS

Simplcity Robotics, Research Intern Dec. 2025 – Present

- Research on embodied intelligence and generalizable robot foundation models.

Beijing Academy of Artificial Intelligence (BAAI), Research Intern Aug. 2024 – Oct. 2025

- Research on embodied-brain benchmarks for multimodal large language models and robot intelligence.

ByteDance, Research Intern

Mar. 2024 – Aug. 2024

- Trust & Safety Applied Algorithm Team for Ads and E-commerce.

HONORS AND AWARDS

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|---|------------------|
| • Shanghai Jiao Tong University Outstanding Graduate | Jun. 2023 |
| • Weichai Power Scholarship | Oct. 2022 |
| • Huawei Scholarship | Dec. 2021 |
| • Shanghai Jiao Tong University B-Class Merit Scholarship | 2020, 2021, 2022 |

ACADEMIC SERVICE

- Reviewer for AAAI 2026, ICLR 2026, CVPR 2026, ECCV 2026, and NeurIPS 2026 Datasets & Benchmarks.